Supplementary Information

Few biological parameters suffice to reproduce measured dynamics in systems biology models: local sensitivity-based reduction with formally verified mathematical guarantees

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Supplementary Table S1. Selected biological parameters

Final subset S for each problem (after backward elimination) and the parameters removed by backward elimination. Calibration parameters, which are always re-estimated, are not listed. Crauste and Beer have no admissible subset; the subset at which the method stopped is shown. For Crauste it is not meaningful: about half of the perturbed simulations have no solution on the measured horizon (finite-time growth without bound), so the sensitivity matrix is incomplete.

System	Selected biological parameters S	Removed by backward elimination
Alkan	kt, k_rep_nhej	–
Armistead	k_d, alpha_cer	k00, k2
Bachmann	CISRNADelay, JAK2ActEpo	–
Beer	kdegi_typeIDT1, kdegi_typeIDT3, ksyn_typeIDT3, kdim_typeIDT1, beta_typeIDT3_ExpID2, beta_typeIDT3_ExpID3, beta_typeIDT3_ExpID5, beta_typeIDT3_ExpID1, beta_typeIDT3_ExpID4, init_Bac, beta_typeIDT1_ExpID6, beta_typeIDT1_ExpID5, beta_typeIDT1_ExpID1, beta_typeIDT1_ExpID4, beta_typeIDT1_ExpID3, kdim_typeIDwt, ksyn_typeIDT5, kdim_typeIDT5	–
Bertozzi	R0_NY, I0_NY	–
Blasi	a_basal, a_k16_k12k16	a_k12k16_k8k12k16, a_k5k12_k5k8k12, a_k8k12k16_4ac
Boehm	k_imp_hetero, k_phos	–
Borghans	Kz, K_par	–
Brannmark	k1f, km2	–
Bruno	kb1, init_bcar2, init_zea_1, init_bcar1, init_bcry_1	szea, init_b10_1
Chen	k106, k44	–
Crauste	rho_E	–
Elowitz	n_Hill, tau_mRNA	tau_prot, tau_prot_GFP, eff
Fiedler	tau2, k6	–
Fujita	init_AKT, reaction_6_k1	–
Giordano	alpha_4, alpha_28	–
Isensee	kf_RIIp_C_2__RII_C_2, xi_kf_RII_C_2__RII_2	ks_AC_cAMP, KD_cAMP, xi_AC_cAMP_Fsk, kf_RIIp_cAMP_C_2__RIIp_2, kf_RII_2__RII_C_2
Lang	kDeE2f1, kDeCe	–
Laske	k_imp, k_syn_R_M	k_bind_M1
Liu	kappa, phi	–

System	Selected biological parameters S	Removed by backward elimination
Lucarelli	init_Rec, S2tot, S3tot, S_dephosphos, S_phos, geneF_turn, k_234, geneK_turn, S4tot, geneH_turn	–
Okuonghae	transmission_rate_effective, gamma_i	–
Oliveira	beta_0, h_hosp_rate	–
Perelson	c, delta	–
Rahman	infected_moderate_transmission_rate, infected_normal_worsen_rate	–
Raia	SOCS3mRNA_production, init_Rec_i	–
Raimundez	kdeg_membran__MKN1, d_kimp_pEGFR_EGF_2__MKN1_2_HS746T	ka_MAPK__MKN1
SalazarCavazos	ratio_kpkd_Y1068__FREE, ratio_kpkd_YN__FREE	–
Schwen	ini_R1, ini_R2fold	–
Smith	k8, kminus7b	–
Sneyd	l4, l2	–
Weber	s12, a21	–
Zhao	R_Stage_I_China, R_Stage_II_Wuhan, gamma_1_Stage_I_Hubei	–
Zheng	k01_02, k13_12	–

Supplementary Table S2. Notation

Symbol	Meaning	Value used
$x(t;\theta) \in \mathbb{R}^n$	state trajectory of the model $\dot{x} = f(x, \theta)$	–
$\theta \in \mathbb{R}^p$, θ_0	parameter vector; nominal (published) value	PEtab nominal values
Θ_B, Θ_C	biological parameters (in the SBML model) and calibration parameters (observation scalings and offsets)	from the PEtab tables
$y_i(\theta) = h(g_i(x(t_i), \theta))/\sigma_i$	i-th scaled output: observable g_i, transformation h (lin, log, $\log_{10}$), noise s.d. σ_i, at the condition and time of measurement i	PEtab measurement table
$J = \partial y / \partial \log \theta \in \mathbb{R}^{m \times p}$	relative sensitivity matrix (central differences, step δ)	$\delta = 0.01$
$E_j = \|J_{:,j}\|^2$	sensitivity energy of parameter j	–
$R_{\text{var}}(S)$	fraction of the energy of biological parameters captured by S	$\tau_R = 0.89$
s_j	finite-perturbation effect of parameter j (stage-2 ordering)	$\eta = 0.1$
$\kappa(S)$, $\text{VIF}(S)$	condition number and largest variance inflation factor of the normalised columns of J_S	$\tau_\kappa = \tau_{\text{VIF}} = 10$
Δy_{full}, Δy_S	output change when all parameters, or only those in $S \cup \Theta_C$, take their scenario values	$N = 15$, $\varepsilon = 0.05$
$\cos \Delta$	cosine similarity between Δy_{full} and Δy_S	–
e_{rel}	relative error without re-estimation	–

Symbol	Meaning	Value used
e_{fit}	relative error after re-estimating θ_S and θ_C	–
$\tau_e = \sqrt{1-c^2}$	admissibility threshold on the median e_{fit}	$c = 0.9$, $\tau_e = 0.436$
$k_{\min}$	minimum subset size	2
χ^2/n	mean squared noise-scaled residual on test data	–
e_{pred}	relative prediction error with respect to a known truth (synthetic experiment)	–

Supplementary Table S3. Formal proofs

All formal proofs are collected in the folder `lean_proofs/` of the repository, a self-contained Lean 4 project (Lean 4.23.0, Mathlib v4.23.0) that compiles with a single command (`./verify.sh`). It contains 31 theory modules with 289 theorems and lemmas, and 22 benchmark models translated from SBML. No file contains `sorry` or `axiom`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for every theorem. The file `THEOREMS.md` in that folder lists every theorem and lemma; n is the number of theorems and lemmas per module.

Module	n	Content	Main theorems
Sensitivity energy and trajectory error			
`FirstOrderErrorBound`	12	Energy $E_j = (J^\top J)_{jj}$, R_{var} and first-order output error when parameters are fixed	`total_eq_kept_add_removed`, `first_order_error_from_Rvar`
`TrajectoryErrorBound`	9	From the Jacobian to the trajectory: error bound when discarded parameters are fixed	`sqrt_removedParamSq_le`
`CertifiedODE Reduction`	1	Local certificate: $R_{\text{var}}(S) \geq r$ bounds the trajectory error of the reduced model	`certified_parameter_reduction`
Differential equations: existence and sensitivities			
`GlobalLipschitzODE`	3	Global existence on $[a, b]$ for globally Lipschitz vector fields	–
`ParamDiffODE`	8	Solutions depend differentiably on parameters and initial values; the derivative solves the variational equation	`hasFDerivAt_solution_param_init`
`LocalExistence`	9	Solutions exist for parameters near θ_0 when the nominal solution exists	–
`SensitivityLipschitz`	8	Explicit Lipschitz constants of trajectories and sensitivities in θ (Grönwall)	`sens_lipschitz`, `traj_lipschitz`
Finite perturbations and cosine certificates			
`CosineCertificate`	5	Non-asymptotic bounds for finite perturbations; per-scenario certificate of $\cos\Delta$	`finite_trajectory_error`, `finite_cos_certificate`
`CertifiedFiniteODE`	5	Certificate of $\cos\Delta$ for the ODE with an explicitly bounded Lipschitz constant	`certified_cos_ode`
`ExactLinear Certificate`	13	Certificates with the exact linear error; local version without Lipschitz constants	`finite_cos_certificate_exact`, `certified_cos_ode_exact`
`NominalCertificate`	1	Local $\cos\Delta$ certificate from the nominal trajectory only	–

Module	n	Content	Main theorems
`ScaledError`	5	Theorem 1: the minimum relative error after rescaling equals $\sqrt{1-\cos^2\Delta}$	`min_rel_error_eq`
Selection, conditioning and identifiability			
`GreedySelection`	7	Greedy selection returns a subset with $\kappa \le \tau_\kappa$ and VIF $\le \tau_{\mathrm{VIF}}$; permanent discard is correct	`greedy_kappa_vif_guarantee`
`Identifiability` `Conditioning`	10	κ and VIF bounds imply unique and stable least-squares estimates	`ls_unique, ls_param_kappa_stability, ls_vif_inverse_stability`
`SpectralConditioning`	22	κ and VIF exactly as computed numerically (spectral theorem, matrix inverse)	–
`NumericalRobustness`	6	Finite-difference error; R_{var}, κ and VIF certified from approximate values with margins	`fd_column_error, rvar_certified_from_approx`
`PatternMechanisms`	4	Opposing effects imply rejection by κ and non-identifiability; zero discarded energy implies exact first-order reduction	`not_okKappa_of_kernel, exact_reduction_of_zero_energy`
Re-estimation, minimality and measured outputs			
`RefitMonotone`	5	Re-estimating more parameters never increases the fitting error; $e_{\mathrm{fit}} \le e_{\mathrm{rel}}$	`fitErr_anti, fitErr_le_erel, admissible_mono`
`Poda`	2	Theorem 2: backward elimination returns an inclusion-minimal admissible subset	`prune_spec, exists_minimal_admissible`
`OutputComposition`	4	Error bounds transfer from states to measured outputs, including log and $\log_{10}$	`output_uniform_bound, log10_lipschitz`
`LinearizedFit`	3	The first-order e_{fit} differs from the exact one by a second-order term	`lin_le_scaled, fit_lower_of_lin`
Model assumptions for kinetic models			
`KineticRegularity`	11	Rate laws (mass action, Michaelis–Menten, Hill, ...) define smooth vector fields on their domain	–
`EventSystems`	1	Systems with events at fixed times: existence and differentiability across segments	–
`PositivityInvariance`	15	Quasi-positive models keep concentrations non-negative	–
`KineticCheck`	19	Verified automatic checkers for translated rate laws	–
`GlobalExistence`	11	Global existence under linear growth on the non-negative orthant	–
`KineticNetwork`	18	Reaction networks: complete check including global existence	`network_final`
`StrictExistence`	12	Global existence with strictly positive concentrations	`exists_global_solution_strict`
`StrictNetwork`	46	Networks with strictly positive domains (e.g. $\beta SI/N$, Hill with real exponent)	`strict_final_init`
`RiccatiNetwork`	9	Existence on a finite horizon under quadratic growth (Riccati bound; used for Crauste)	`riccati_final`
`IntervalInit`	5	Parameter-dependent initial values certified with rational interval arithmetic	`strict_final_initI`

Module	n	Content	Main theorems
Total	289		

Verified benchmark models. Armistead, Bachmann, Blasi, Boehm, Borghans, Brannmark, Chen (500 species, in four parts), Crauste (horizon $T \leq 1$), Elowitz, Fiedler, Froehlich (1,228 species, in twelve parts), Giordano, Lang, Okuonghae, Rahman, Raia, Raimundez, SalazarCavazos, Sneyd, Weber, Zhao and Zheng (folder `lean_proofs/Models/`). For each model, the final theorem establishes that the vector field is smooth on its domain, that the solution exists on the simulated horizon and that concentrations remain non-negative.

Supplementary Note. Verification of the simulator

Our simulation of the measured outputs was compared, at the nominal parameters, with the reference simulations distributed with the benchmark, matching each measurement by observable, condition and time. For the problems with comparable reference files the median relative difference was below 10^{-5} in all but five cases: Chen (1.1×10^{-4}), Zheng (whose reference file contains the measured data rather than a simulation), Crauste (0.8%, very stiff model), Lang (2.8%, an oscillating cell-cycle model simulated to $t = 136{,}472$, where peaks differ by an accumulated phase shift) and Perelson (17%; the initial state of the model is stationary and our simulation respects it, which suggests that the reference was generated with different values). Several problems have no comparable reference (no file, or different condition identifiers). The full comparison is given in the repository (`certificados/resultados/salidas/VERIFICACION_SIMULADOR.md`).